

Ch12 13 Fixed Floating Point Number Quiz

1. In the $Q_m.n$ fixed-point notation (signed), what is the total number of bits used to represent the number?

- A) $m + n$
- B) $m + n + 1$
- C) $m \times n$
- D) $m - n$

ANS:

2. How is the value of a signed fixed-point number ($Q_m.n$) calculated using the weighted bit method (where the MSB is the sign bit)?

- A) The sign bit has a weight of $-2^{(m+1)}$.
- B) The sign bit has a weight of -2^m .
- C) The sign bit is ignored in calculation.
- D) The sign bit has a positive weight of 2^m .

ANS:

3. In IEEE 754 Single Precision (FP32) format, what is the "Bias" value used for the exponent?

- A) 127
- B) 128
- C) 255
- D) 1023

ANS:

4. Which of the following bit patterns represents the value "1.0" in IEEE 754 Single Precision format (Hexadecimal: 0x3F800000)?

- A) Sign=0, Exponent=128, Fraction=0
- B) Sign=1, Exponent=127, Fraction=0
- C) Sign=0, Exponent=127, Fraction=0
- D) Sign=0, Exponent=0, Fraction=1

ANS:

5. When converting a floating-point number f to a fixed-point number with n fractional bits, what is the first step in the calculation?

- A) Divide f by 2^n .
- B) Add the bias to f .
- C) Multiply f by 2^n .
- D) Invert the bits of f .

ANS:

6. In IEEE 754, what does an exponent field of all 1s (e.g., 255 for FP32) and a non-zero fraction field represent?

- A) Positive Infinity ($+\infty$)
- B) Negative Infinity ($-\infty$)
- C) Zero
- D) NaN (Not a Number)

ANS:

7. What is the primary difference between "Normalized" and "Subnormalized" numbers in IEEE 754?

- A) Subnormalized numbers have an implicit leading bit of 0, while Normalized numbers have an implicit 1.
- B) Subnormalized numbers use a bias of 255.

- C) Normalized numbers cannot represent negative values.
- D) Subnormalized numbers are used only for values greater than 1.

ANS:

8. According to the document, how does Fixed-Point representation differ from Floating-Point regarding precision?

- A) Fixed-Point precision decreases as the number's magnitude increases.
- B) Floating-Point numbers are evenly distributed across the entire range.
- C) Fixed-Point numbers are evenly distributed, meaning precision is fixed.
- D) Fixed-Point generally has a much larger range than Floating-Point.

ANS:

9. What are the "bf16" and "bf8" formats primarily used for?

- A) High-precision scientific simulations requiring 64-bit accuracy.
- B) AI and machine learning to trade slightly lower accuracy for speed and efficiency.
- C) Legacy banking systems using COBOL.
- D) Storing high-fidelity audio files.

ANS:

10. If you decode the binary sequence 1 10000011 1111... (Sign=1, Exponent=131), what is the resulting sign and exponent value before bias adjustment?

- A) Sign: Positive, Exponent: 4
- B) Sign: Negative, Exponent: 131 (Actual exponent $131 - 127 = 4$)
- C) Sign: Negative, Exponent: 3
- D) Sign: Positive, Exponent: -4

ANS: